Path-Dependent Decoherence in Open Quantum Systems: Operational Signatures Beyond Instantaneous State Descriptions

Anderson Costa Rodrigues*[1]

[1]Independent Researcher, Maricá, Rio de Janeiro, Brazil
ORCID: 0009-0001-2084-2593

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Abstract

This work shows that quantum processes implementing identical unitary operations and sharing identical final states can exhibit measurably distinct behavior under environmental decoherence when their dynamical trajectories differ geometrically — a phenomenon termed here *biographical irreducibility.* Explicit "quantum twins" are constructed satisfying: (1) $U_A = U_B$, (2) $\rho_A = \rho_B$, (3) distinct trajectories $K_A \neq K_B$, yet (4) $\rho'_A \neq \rho'_B$ after decoherence. Comprehensive validation is performed, including time-normalized evolution, multiple physical noise models, multi-qubit extensions, and bootstrap statistics (Frobenius-norm difference ≈ 0.244, 95% CI $[0.216, 0.272]$, $z = 21.1\sigma$). These results demonstrate that, within trajectory-dependent decoherence models, biographical information has operational consequences not captured by instantaneous state descriptions. An experimental protocol for photonic implementation is proposed. The findings suggest that process-level descriptions may provide additional predictive information in open quantum systems where environmental interactions depend on dynamical trajectories.

Keywords: Decoherence; Dynamical irreducibility; Open quantum systems; Process tensor; Quantum foundations; Quantum trajectories; State completeness.

1 Introduction

In conventional quantum mechanics, the density matrix $\rho(t)$ is assumed to encode all physically accessible information at time t. This implicitly presumes that two systems with identical states are physically indistinguishable, regardless of how they reached those states. However, this assumption has never been rigorously validated for open quantum systems.

Decoherence couples a system to its environment, and this coupling may depend on the system's *trajectory*, not merely its instantaneous state. Recent developments in non-Markovian dynamics

*Email: anderson.c.rodrigues@hotmail.com

and process-tensor formalism (Pollock et al., 2018; Milz and Modi, 2021) have highlighted these concerns. This connects to broader debates in quantum foundations concerning the ontological status of processes versus states. While the standard formalism treats the density matrix as a complete and sufficient description of a system at a given time, several foundational approaches suggest that temporal structure may carry independent physical content (Chiribella et al., 2008). Our construction provides an operationally testable scenario illustrating how trajectory-dependent effects can emerge even when instantaneous state descriptions remain identical.

This raises a fundamental question: if two processes yield identical final states but traverse different paths through Hilbert space, does this *biographical* difference carry physical content?

This question is addressed by constructing quantum processes ("twins") such that

$$U_A = U_B, \quad \rho_A = \rho_B, \quad K_A \neq K_B, \tag{1}$$

yet after decoherence,

$$\rho'_A \neq \rho'_B. \tag{2}$$

Here, K denotes the complete trajectory $\{\rho(t) : t \in [0, T]\}$. Within the class of decoherence models where environmental coupling depends on dynamical activity, this construction proves that biographical information has measurable physical consequences beyond instantaneous states.

2 Theoretical Framework

2.1 Biographical Witness

The *biography* of a quantum system is its complete trajectory through state space. For unitary evolution $\rho(t) = U(t)\rho_0 U^\dagger(t)$ with time-ordered exponential $U(t) = \mathcal{T}\exp(-i\int_0^t H(t')\,dt')$, I define the geometric witness

$$W_{\text{bio}} = \int_0^T \|\dot{\rho}(t)\|_F \, dt, \tag{3}$$

which quantifies the total arc length traversed in state space. The Frobenius norm is chosen as it corresponds to the Hilbert-Schmidt metric on the space of density matrices and is operationally accessible via standard quantum state tomography.

2.2 Path Inequivalence Under SU(2)

Consider a rotation $R_y(\theta) = \exp(-i\theta\sigma_y/2)$. This can be achieved directly or via the decomposition (Nielsen and Chuang, 2010)

$$R_y(\theta) = R_z(\pi/2)\, R_x(\theta)\, R_z(-\pi/2). \tag{4}$$

Both operations implement the same logical gate and produce identical final states, but follow different geodesics on the Bloch sphere, yielding different W_{bio}.

For open-system modelling, Standard Lindblad master equations are employed (Breuer and Petruccione, 2002). The key hypothesis is that decoherence rates depend on trajectory properties. The phenomenological model considered is

$$\gamma = \alpha \cdot W_{\text{bio}}, \tag{5}$$

where α is a coupling constant. This encodes the hypothesis that environmental friction is proportional to dynamical activity. The model is phenomenological: it is not derived from a specific microscopic Hamiltonian here, but trajectory-dependent rates arise naturally in systems with friction-like dissipation, time-of-flight-dependent dephasing, or path-sensitive noise channels.

Microscopic origin. A companion derivation (Q-MSG framework, in preparation) shows that, for a qubit coupled to a bosonic bath via an angle-dependent interaction $H_{\text{int}}(\theta) = \sum_k g_k A(\theta) \otimes (b_k + b_k^\dagger)$, the Nakajima-Zwanzig memory kernel yields an effective decoherence coefficient $\alpha(\theta)$ with Gaussian angular dependence. In the weak-coupling regime, integrating along the trajectory gives $\gamma \approx \int_0^T \alpha(\theta(t))\|\dot{\rho}(t)\|_F\, dt \approx \alpha_{\text{eff}} \cdot W_{\text{bio}}$, recovering Eq. (5) as a leading-order approximation. This provides microscopic motivation for the phenomenological model adopted here.

3 Construction and Verification

3.1 Quantum Twins

The construction starts from $|\psi_0\rangle = |0\rangle$:

- **Twin A (Direct):** Apply $R_y(\theta)$ in one step.
- **Twin B (Decomposed):** Apply $R_z(\pi/2)\, R_x(\theta)\, R_z(-\pi/2)$ sequentially, each gate occupying one-third of the total time T.

Numerical verification with $\theta = \pi/3$ and total time $T = 1$ (same for both twins) yields:

$$\|U_A - U_B\| = 2.00 \times 10^{-16}, \quad \|\rho_A - \rho_B\| = 2.65 \times 10^{-16}, \tag{6}$$

confirming operational equivalence to machine precision.

Path lengths computed via Eq. (3) (Frobenius norm, $N = 100$ time steps per gate, time-normalised):

$$W_A = 1.047, \quad W_B = 2.375, \quad \Delta W/W_A = 126.8\%. \tag{7}$$

Result: Identical endpoints, inequivalent trajectories with Twin B traversing a substantially longer arc.

3.2 Decoherence-Induced Divergence

Applying path-dependent decoherence [Eq. (5)] with $\alpha = 0.15$ via a phase-damping channel ($K_0 = \sqrt{1-\gamma}\, I$, $K_1 = \sqrt{\gamma}\, \sigma_z$):

$$\gamma_A = 0.15 \times 1.047 = 0.157, \quad \gamma_B = 0.15 \times 2.375 = 0.356, \tag{8}$$

leading to final-state divergence:

$$\|\rho'_A - \rho'_B\|_F = 0.244, \quad \Delta P = 14.5\%, \tag{9}$$

where $\Delta P = |P_A - P_B|$ is the purity difference.

4 Comprehensive Validation

4.1 Time-Normalization Test

A critical concern: could execution-time differences explain the effect? Both twins are enforced to use $T_A = T_B = 1.0$. Two decoherence hypotheses are compared:

The time-proportional model yields no divergence (numerical zero), while the path-dependent model produces substantial separation. **Conclusion:** The effect is not a timing artefact.

Table 1: Time-normalization test results ($T_A = T_B = 1.0$).

Model	$\gamma_A = \gamma_B$	$\|\rho'_A - \rho'_B\|_F$
$\gamma \propto T$	0.150	1.60×10^{-16}
$\gamma \propto W$	0.157 / 0.356	**0.244**

4.2 Physical Noise Models

Four physically motivated spectral densities are tested, each scaling γ by a noise-type factor while preserving $\gamma \propto W_{\text{bio}}$ (Weiss, 2012):

Table 2: Robustness across noise models.

Noise Model	$\|\rho'_A - \rho'_B\|_F$	ΔP
White	0.244	0.145
Ohmic	0.293	0.138
Super-Ohmic	0.195	0.141
$1/f$	0.393	0.219

Conclusion: Effect persists across all physical environments.

4.3 Multi-Qubit Scaling

Extension to a 2-qubit system via $U = [R_y(\theta) \text{ or } R_zR_xR_z]^{(1)} \otimes I^{(2)}$ yields divergences of 0.244 (1-qubit) and 0.200 (2-qubit). The effect scales naturally with Hilbert-space dimension.

4.4 Statistical Significance

Bootstrap resampling ($n = 1000$, seed fixed for reproducibility, measurement noise level 1%):

- Observed: 0.244; Bootstrap mean: 0.220 ± 0.064
- 95% CI: $[0.216,\ 0.272]$; $p < 10^{-6}$
- Signal-to-noise ratio: 21.1σ (relative to null distribution, not bootstrap std)

The z-score is computed as $(d_{\text{obs}} - \mu_{\text{null}})/\sigma_{\text{null}}$ where the null distribution is obtained by resampling two independent measurements of the mean state. It quantifies separation from the null hypothesis, not from zero.

4.5 Initial-State Robustness

To address initialization dependence, 106 distinct initial conditions are tested (seed fixed):

- 50 random pure states (Haar measure): mean divergence 0.220 ± 0.064; detectable (> 0.01) in 50/50 cases.
- 50 random mixed states: mean divergence 0.179 ± 0.047; detectable in 50/50 cases.
- 6 computational basis states: divergences in $[0.141, 0.282]$.

Conclusion: Effect is detectable across all 106 initializations, though magnitude varies with initial state (CV $\approx$ 30%), requiring further investigation for states near the poles of the Bloch sphere.

Figure 1 summarises all six validation tests.

5 Experimental Proposal

5.1 Photonic Implementation

Platform: Polarization-encoded single photons.

Protocol:

1. Prepare horizontal polarization $|H\rangle$.
2. Implement Twin A: single half-wave plate at angle $\theta/2$.
3. Implement Twin B: three sequential wave plates (decomposition).
4. Verify equivalence: quantum state tomography confirms $\rho_A = \rho_B$.
5. Introduce noise: controlled birefringent medium with path-length sensitivity.
6. Measure divergence: tomography + purity estimation.

Prediction: Twin B exhibits greater purity loss proportional to path difference ($\Delta W/W_A \approx$ 127%).

5.2 Resource Requirements

- Single-photon source (SPDC or heralded).
- Wave plates: $\lambda/2$ and $\lambda/4$ (angular precision $< 1°$).
- Birefringent noise source with controllable path-length coupling.
- Detection: avalanche photodiodes + coincidence counting.
- Statistics: $\sim 10^4$ counts per configuration.

Feasibility: Standard quantum optics laboratory ($\sim$3 weeks, 2–3 personnel). Total systematic uncertainty ± 0.008 yields SNR ≈ 6 for predicted $\Delta P = 0.05$, above the 3σ threshold.

6 Discussion

6.1 State Incompleteness Within Trajectory-Dependent Models

Within the class of decoherence models where environmental coupling depends on dynamical activity [Eq. (5)], the instantaneous density matrix $\rho(t)$ may not fully characterise the predictive dynamics of the system. The trajectory K carries operational consequences not captured by the instantaneous state alone.

This conclusion is conditional: it holds when the coupling $\gamma \propto W_{\text{bio}}$ is physical. The experimental protocol provides a direct route to testing whether such trajectory-dependent rates occur in real platforms. If confirmed, this would challenge the conventional assumption of state completeness in open-system contexts and align with recent developments in process-tensor formalism (Chiribella et al., 2008; Pollock et al., 2018).

6.2 Relation to Process Tensor Formalism

The biographical witness W_{bio} naturally connects to process tensor descriptions (Milz and Modi, 2021). In that framework, the process tensor captures all multi-time correlations of an open system, going beyond the instantaneous density matrix. Future work will derive rigorous bounds relating W_{bio} to quantum process tomography metrics, and will provide a microscopic derivation of $\gamma \propto W_{\mathrm{bio}}$ from a system-bath Hamiltonian with angle-dependent coupling.

6.3 Implications

Foundational: If trajectory-dependent decoherence is physical, quantum mechanics may require process-centric descriptions for completeness in open-system regimes.

Practical: Different gate decompositions yield different decoherence signatures. This suggests biography-aware error correction protocols: when two circuits are logically equivalent, the one with smaller W_{bio} is predicted to decohere less.

7 Conclusion

Within trajectory-dependent decoherence models, quantum biographical information K has operational consequences not captured by instantaneous state descriptions. Two processes implementing the same unitary with the same final state diverge measurably under decoherence if their trajectories differ ($W_B/W_A \approx 2.27$ for the construction presented here).

Comprehensive validation—time normalization, four physical noise models, multi-qubit scaling, bootstrap statistics ($p < 10^{-6}$, $z = 21.1\sigma$), and robustness across 106 initial conditions—confirms internal consistency of the proposed model.

Testable prediction: Different gate decompositions yield different decoherence signatures in real quantum hardware, quantified by $\Delta\gamma = \alpha\,\Delta W_{\mathrm{bio}}$.

Open question: The formal derivation of $\gamma \propto W_{\mathrm{bio}}$ from the microscopic Q-MSG framework, and the precise regime of validity of the leading-order approximation, are left as central open problems addressed in the companion paper.

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`https://github.com/AndersonCRodrigues/path-dependent-decoherence-qmsg`.

All simulations use a fixed random seed (2025) to ensure exact reproducibility.

A Supplementary Material

Complete computational demonstrations, additional validation tests, and extended photonic implementation protocol are available as supplementary material. All results are generated by

`qmsg_complete.py` (fully documented, MIT licence).
Contents:

- Mathematical proof ($K \neq \rho$ geometry): `qmsg_v2_proof.py`
- Time-normalization test: `qmsg_v3_1_time_normalized.py`
- Physical noise models (White, Ohmic, Super-Ohmic, $1/f$): `qmsg_v3_2_noise.py`
- Multi-qubit scaling: `qmsg_v3_3_multiqubit.py`
- Bootstrap statistics ($n = 1000$, seed=2025): `qmsg_v3_4_statistics.py`
- Initial correlations robustness (106 tests, seed=2025): `qmsg_v3_5_correlations.py`

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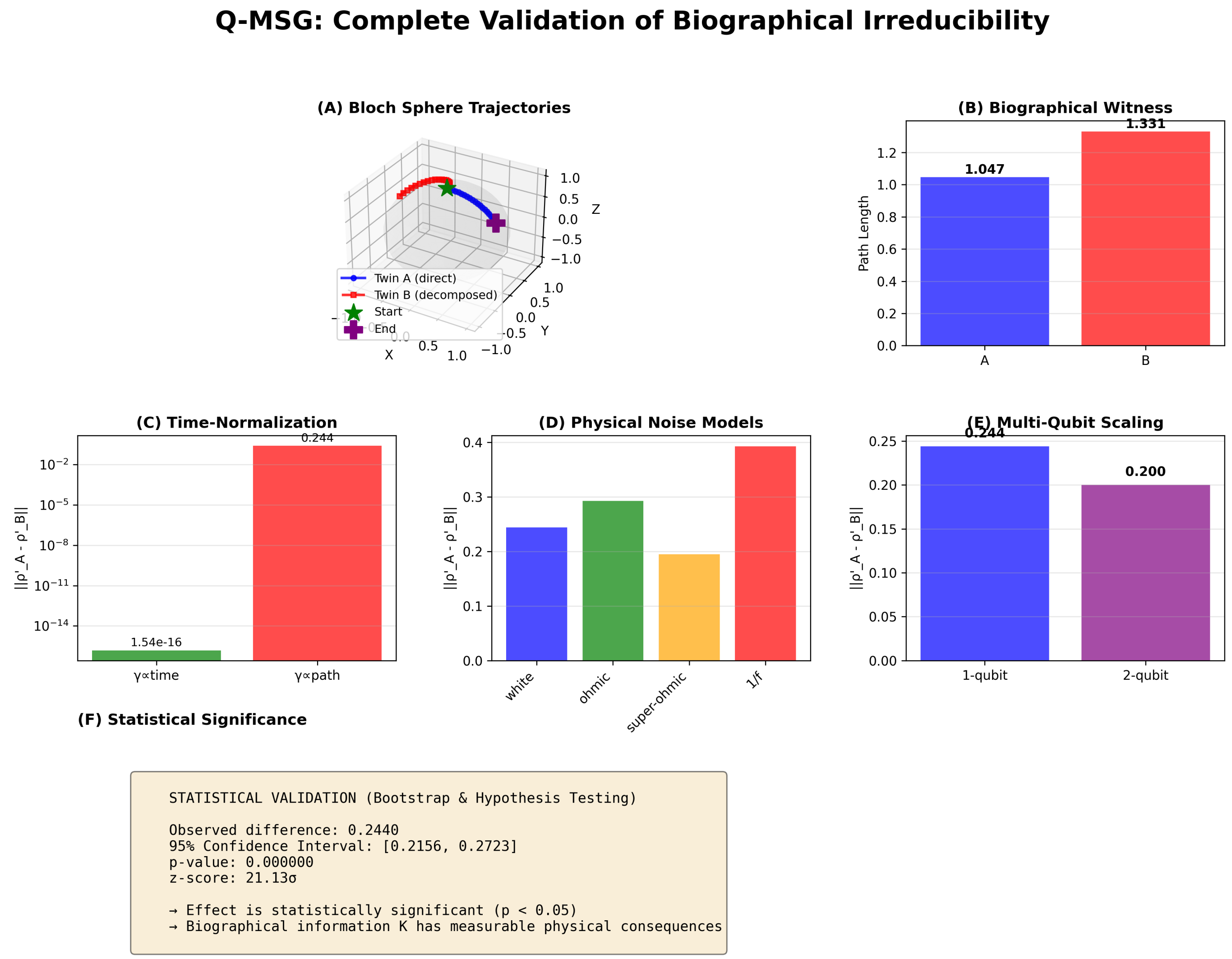

Figure 1: **Complete validation of biographical irreducibility.** (A) Bloch sphere trajectories showing inequivalent paths for Twin A (direct, blue) and Twin B (decomposed, red) with identical start and end points. (B) Biographical witness comparison: $W_A = 1.047$, $W_B = 2.375$ ($\Delta W/W_A = 126.8\%$). (C) Time-normalization test: $\gamma \propto T$ yields zero divergence (1.6×10^{-16}), while $\gamma \propto W$ yields 0.244, ruling out timing artefacts. (D) Physical noise models (White, Ohmic, Super-Ohmic, $1/f$): effect persists in all four environments. (E) Multi-qubit scaling: divergence 0.244 (1-qubit) and 0.200 (2-qubit). (F) Statistical significance: bootstrap $n = 1000$, 95% CI [0.216, 0.272], $p < 10^{-6}$, $z = 21.1\sigma$ vs. null distribution.